Conditional resampling and retained boundary entropy in finite SU(2) models

Project Stardust

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Abstract

A conditional update that retains its boundary can remove all error in the conditional law without removing any error in the boundary marginal. We give complete probability-level proofs of the update law, idempotence, stationary family and exact finite relative-entropy decomposition. For bounded finite-word SU(2) interactions we construct the globally normalized joint law and its actual conditional kernel, including the interaction-induced boundary marginal. Conditioning that continuous boundary on a nontrivial open hemisphere gives a stationary perturbation with finite strictly positive entropy against the original joint law. Every strict global entropy contraction factor therefore fails for this update, at every fixed finite coupling. The obstruction uses the original continuous-boundary reference, in addition to an elementary two-atom example. A separate state-replacement loop kernel has genuine fixed-coupling entropy decay, with finite entropy after one step even from a point mass. A zero-trace band supplies a genuine two-step Haar reset with explicit pair-count bounds; an exact zero-trace row separates this mechanism from Wilson transport. Classical conditional-expectation and heat-bath mechanisms are credited; no obstruction to dynamics updating all coordinates or to a physical Yang–Mills mass gap is asserted.

1 The distinction to be proved

Fix a boundary variable p and an interior variable x. A heat-bath step retains p and replaces x by a draw from a specified conditional law K_p. Thus it enforces the desired conditional distribution in one step. It does not force the boundary marginal to equal any chosen reference marginal. This elementary fact becomes useful when an entropy chain rule is mistaken for a strictly positive global dissipation estimate.

The frozen verified source for this exposition is commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The accompanying map supplies exact declaration headers, hashes and links. The main construction is already in that library; no new research proof, mathematical source change or external submission is made here. We work with probability measures on standard Borel spaces, sufficient for all continuous compact-group examples. The formal generic kernel results also cover more general measurable spaces and, in their stated forms, finite or sigma-finite measures. We do not identify those broader hypotheses with the narrower probability-level exposition.

For probability measures P, Q, relative entropy means

$$\mathrm{D}(P\|Q) = \int f \log f \, dQ \quad \text{if } P = fQ, \qquad \mathrm{D}(P\|Q) = +\infty \quad \text{if } P \not\ll Q.$$

Use $0 \log 0 = 0$. This is an extended nonnegative quantity. The integral is well defined because the negative part of $f \log f$ is bounded by $1/e$. For the exact decomposition below we assume finite input entropy, so all terms can be handled without subtracting infinities. The actual SU(2) counterexample will have entropy equal to a specified finite positive number.

2 An actual boundary-preserving kernel

Let P, E be standard Borel spaces, and let $K : \mathsf{P} \to \mathcal{P}(\mathsf{E})$ be a Markov kernel. Write $\pi \otimes K$ for the joint measure

$$(\pi \otimes K)(B) = \int\int 1_B(p, x)\, K_p(dx)\, \pi(dp).$$

Define a kernel T on $\mathsf{P} \times \mathsf{E}$ by

$$T((p, x), B) = \int 1_B(p, y)\, K_p(dy). \tag{1}$$

For a law J, denote its boundary marginal by J_{P} and its updated law by JT. This notation corresponds to kernel composition in the formal sources, whose composition symbol is written in the opposite order.

Theorem 2.1 (Update law and stationary family)**.** *The kernel T is Markov, and for every probability law J,*

$$JT = J_{\mathsf{P}} \otimes K. \tag{2}$$

It preserves the boundary marginal and satisfies $T^2 = T$. Every law $\pi \otimes K$ is stationary. Conversely, $JT = J$ exactly when $J = J_{\mathsf{P}} \otimes K$, equivalently when K is a conditional kernel for J.

Proof. For each input (p, x), (1) is the pushforward of the probability K_p under $y \mapsto (p, y)$. For each measurable output set B, the kernel integration theorem makes its value measurable in p, and hence in (p, x). This proves that it is Markov. On rectangles $A \times C$, Tonelli gives

$$(JT)(A \times C) = \int 1_A(p) K_p(C)\, J(dp, dx) = \int_A K_p(C)\, J_{\mathsf{P}}(dp).$$

Rectangles determine probability measures on the product sigma algebra, so (2) follows. Taking $C = \mathsf{E}$ proves marginal preservation. Applying the same formula twice gives $(JT)T = (J_{\mathsf{P}} \otimes K)T = J_{\mathsf{P}} \otimes K$. The same rectangle calculation for a point input proves the kernel identity $T^2 = T$. The stationary statements follow directly from (2) and the definition of a conditional kernel. □

Source correspondence: `ConditionalResamplingKernel`
`update_markov; update_law; boundary_preserved; update_idempotent`

Source correspondence: `ConditionalResamplingKernel`
`every_boundary_stationary; invariant_iff_conditional`

Different boundary marginals yield different stationary laws: the first marginal of $\pi \otimes K$ is π. Thus stationary nonuniqueness here is not ambiguity about versions of a conditional distribution. It is a consequence of a coordinate that the update retains exactly.

3 Exact finite-entropy accounting

Fix a reference boundary probability τ and set $Q = \tau \otimes K$. By the preceding theorem, Q is stationary. The following proof explicitly identifies the density after one update.

Theorem 3.1 (Marginal and conditional entropy)**.** *Suppose* $\mathrm{D}(J\|Q) < \infty$*. Then*

$$\mathrm{D}(JT\|Q) = \mathrm{D}(J_{\mathsf{P}}\|\tau), \tag{3}$$

$$\mathrm{D}(J\|Q) = \mathrm{D}(JT\|Q) + \mathrm{D}(J\|JT). \tag{4}$$

All three quantities in (4) *are finite and nonnegative.*

Proof. Finite input entropy implies $J = fQ$ for a nonnegative density f with $\int f\, dQ = 1$. Define

$$m(p) = \int f(p,x)\, K_p(dx).$$

Tonelli gives $J_{\mathsf{P}} = m\tau$ and hence, by (2), $JT = mQ$. Conditional Jensen for the convex function $u \log u$ gives

$$\int m \log m\, d\tau \le \int f \log f\, dQ < \infty.$$

Both negative parts are integrable, so the two finite integrals are unambiguous. Moreover

$$\int |\log m|\, dJ = \int m |\log m|\, d\tau < \infty,$$

where expressions on $m = 0$ can be assigned any value because J gives that set zero mass. The density of JT relative to Q is $m(p)$, so its entropy is precisely $\int m \log m\, d\tau$, proving (3).

On $m(p) > 0$, put $r(p,x) = f(p,x)/m(p)$. On $m = 0$, nonnegativity and the definition of m imply $f = 0$ for K_p-almost every x, for τ-almost every such p. Thus $J \ll JT$ and r is a density of J relative to JT. Since $\log f$ is integrable under J by finite entropy, and $\log m$ is integrable under J by the preceding calculation, their difference $\log r$ is integrable under J. Therefore

$$\mathrm{D}(J\|Q) = \int \log f\, dJ = \int \log m\, dJ + \int \log r\, dJ = \mathrm{D}(JT\|Q) + \mathrm{D}(J\|JT).$$

Nonnegativity follows from Jensen, or from $u \log u \ge u - 1$ integrated against a probability reference. The established integrability gives finiteness of the final term as well. □

Source correspondence: `ConditionalResamplingEntropy`
`entropy_after_update; entropy_decomposition_of_conditional`

The mapped source proves an extended-divergence chain rule using the original measure-theoretic KL library. The displayed human proof deliberately proves its finite-entropy probability restriction, which covers the counterexamples and avoids any assertion that $+\infty - (+\infty)$ is defined.

There is a useful direct interpretation of the second term. For $m(p) > 0$, the input conditional law has density $f(p,x)/m(p)$ relative to K_p. Consequently

$$\mathrm{D}(J\|JT) = \int \mathrm{D}(J(dx \mid p)\|K_p)\, J_{\mathsf{P}}(dp).$$

This is the conditional error that the step removes. The remaining entropy depends only on the retained boundary. On the restricted class $J_{\mathsf{P}} = \tau$, one step does send every input to Q. That restricted statement does not imply strict global contraction for arbitrary inputs.

4 Stationary perturbations with positive finite entropy

The elementary mechanism can be realized without any singular or infinite entropy comparison.

Lemma 4.1 (Conditioning a boundary event)**.** *Let $A \subseteq \mathsf{P}$ be measurable with $a = \tau(A) \in (0,1)$. Define*

$$\pi = \tau(\,\cdot \mid A) = a^{-1} 1_A \tau, \qquad J_A = \pi \otimes K.$$

Then $J_A T = J_A$ and

$$\mathrm{D}(J_A \| Q) = \log(1/a) \in (0, \infty). \tag{5}$$

Proof. Stationarity follows from Theorem 2.1. The density of J_A relative to Q is $a^{-1}1_A(p)$, because integrating on rectangles gives the same joint measure. Its logarithm equals $\log(1/a)$ on the support of J_A, and J_A has mass one. This proves (5) directly. The assumptions $0 < a < 1$ give both finiteness and strict positivity. □

Source correspondence: `ConditioningRelativeEntropy`
`entropy_cond_finite; entropy_cond_exact; entropy_cond_positive`

Theorem 4.2 (No strict global fraction)**.** *Under the assumptions of Lemma 4.1, no $q < 1$ can satisfy*

$$\mathrm{D}(JT \| Q) \le q\, \mathrm{D}(J \| Q)$$

for all finite-entropy probability inputs J. Likewise there is no $c > 0$ such that $c\,\mathrm{D}(J\|Q) \le \mathrm{D}(J\|JT)$ for all such inputs.

Proof. Substitute J_A. The first inequality would say $d \le qd$ for the finite number $d = \log(1/a) > 0$, contradicting $q < 1$. The second would say $cd \le 0$, since $J_A T = J_A$, contradicting $c, d > 0$. □

Source correspondence: `ConditionalResamplingEntropy`
`no_positive_dissipation_fraction; binary_no_strict_contraction`

For a concrete finite example let $\mathsf{P} = \{0,1\}$, $\tau = \frac{1}{2}(\delta_0 + \delta_1)$ and $\pi = \delta_0$. For *any* Markov kernel K, including a strictly positive one on a finite interior space, $\pi \otimes K$ is stationary and its entropy relative to $\tau \otimes K$ is exactly $\log 2$. This example uses a specifically chosen two-atom boundary reference. It does not by itself prove the same statement against a prescribed continuous interaction law; that is the task of the next sections.

5 Globally normalized bounded interaction laws

Let ρ and ν be probability measures on P and E, and let $V : \mathsf{P} \times \mathsf{E} \to \mathbb{R}$ be measurable with $|V(p,x)| \le B < \infty$. Define the section and total partitions

$$z(p) = \int e^{V(p,x)}\, \nu(dx), \qquad \mathcal{Z} = \int z(p)\, \rho(dp).$$

Both satisfy $e^{-B} \le z(p), \mathcal{Z} \le e^{B}$ and are strictly positive. The section partition is measurable by kernel integration. Set

$$K_p(dx) = z(p)^{-1} e^{V(p,x)} \nu(dx), \tag{6}$$

$$\tau(dp) = \mathcal{Z}^{-1} z(p) \rho(dp), \tag{7}$$

$$Q(dp, dx) = \mathcal{Z}^{-1} e^{V(p,x)} \rho(dp) \nu(dx). \tag{8}$$

Proposition 5.1 (Factorization and boundary equivalence)**.** *The formulas define a Markov kernel K and probabilities τ, Q. Moreover $Q = \tau \otimes K$, $Q_{\mathsf{P}} = \tau$, and τ and ρ have exactly the same null sets. In fact*

$$e^{-2B} \le \frac{d\tau}{d\rho}(p) \le e^{2B}.$$

Proof. The section density in (6) is jointly measurable, nonnegative and has integral one. It defines a Markov kernel by the kernel integration theorem. Fubini gives the masses of (7) and (8) as one. Multiplication of the boundary density and conditional density cancels $z(p)$, giving the joint density (8); alternatively integrate on rectangles. The first marginal follows by integrating $K_p(\mathsf{E}) = 1$. The density bound follows from the two partition bounds. A density bounded strictly above zero and finite has precisely the same null sets as its reference measure. □

Source correspondence: `ExponentialJointDisintegration`
`boundary_probability; factorization; joint_probability; conditional_kernel`

Source correspondence: `ExponentialBoundaryEquivalence`
`reverse_ac; null_iff`

The original generic equivalence theorem only requires positive integrable exponential sections and integrability of the joint exponential; the bounded case here supplies these hypotheses directly. The explicit uniform density bound is a human consequence of boundedness, not an asserted additional formal declaration.

The reference boundary is τ, rather than ρ, in the actual globally normalized model. Unless the section partition $z(p)$ is constant, $\rho \otimes K$ differs from Q. Omitting this reweighting would replace the intended reference law and invalidate an entropy comparison claimed to use the globally normalized interaction.

6 Finite SU(2) words and their actual link conditional

Let $G = \mathrm{SU}(2)$, with normalized Haar probability m. Its matrices are unitary of determinant one. For a finite boundary index set I, put $\mathsf{P} = G^I$, $\mathsf{E} = G$, $\rho = m^{\otimes I}$ and $\nu = m$. Choose a finite collection of words $W_j(U)$ in boundary letters U_i and their inverses. Empty words equal the identity. Matrix multiplication and inversion are continuous on G, so each word is continuous, and in particular measurable. Let the collection have N words and define

$$V_\beta(U, A) = \frac{\beta}{2} \operatorname{Re} \operatorname{tr} \left(A \sum_{j=1}^{N} W_j(U) \right). \tag{9}$$

All coupling constants $\beta \in \mathbb{R}$ are allowed.

Lemma 6.1 (Uniform finite-word bound)**.** *For every boundary U and link A, $|V_\beta(U, A)| \le |\beta| N$.*

Proof. Each product $AW_j(U)$ is unitary. Every diagonal entry of a unitary matrix has modulus at most one, since each column has Euclidean norm one. Therefore $|\operatorname{Re} \operatorname{tr}(AW_j(U))| \le 2$. Linearity of the trace and the triangle inequality give a total bound $\frac{|\beta|}{2} \sum_j 2 = |\beta| N$. □

Source correspondence: `SU2BoundaryWordKernel`
`word_measurable; family_measurable; kernel_markov`

Source correspondence: `SU2StapleKernel`

`exponent_bound; exponential_integrable`

The source proves the same bound through its quaternion staple radius; the two-line matrix argument is a human reorganization of that estimate.

Apply Section 5 with $B = |\beta|N$. Denote the resulting actual joint law, boundary marginal and conditional kernel by $Q_\beta, \tau_\beta, K_\beta$. Their explicit densities are

$$Q_\beta(dU, dA) = \frac{e^{V_\beta(U,A)}}{\mathcal{Z}_\beta}\rho(dU)m(dA), \qquad (K_\beta)_U(dA) = \frac{e^{V_\beta(U,A)}}{z_\beta(U)}m(dA).$$

In particular,

$$Q_\beta = \tau_\beta \otimes K_\beta, \qquad \tau_\beta(dU) = \frac{z_\beta(U)}{\mathcal{Z}_\beta}\rho(dU). \tag{10}$$

No formal "conditional density" is being used without a joint law: these formulas construct both the joint probability and its disintegration. The rectangle identity

$$Q_\beta(B \times C) = \int_B (K_\beta)_U(C)\,\tau_\beta(dU)$$

checks the actual conditional relationship. The heat-bath update $T_\beta((U, A), \cdot) = \delta_U \otimes (K_\beta)_U$ is Markov, idempotent, boundary preserving, and leaves Q_β invariant, by Theorem 2.1.

Source correspondence: `SU2BoundaryJointLaw`
`factorization; boundary_marginal; conditional_rectangles; conditional_kernel`

Source correspondence: `SU2ConditionalHeatBath`
`update_markov; update_idempotent; boundary_preserved; joint_invariant`

For every finite-entropy probability input J against this actual joint reference, Theorem 3.1 now gives

$$\begin{aligned} \mathrm{D}(JT_\beta \| Q_\beta) &= \mathrm{D}(J_{\mathsf{P}} \| \tau_\beta), \\ \mathrm{D}(J \| Q_\beta) &= \mathrm{D}(JT_\beta \| Q_\beta) + \mathrm{D}(J \| JT_\beta). \end{aligned}$$

The formal source retains the extended-divergence versions as well.

Source correspondence: `SU2HeatBathEntropy`
`actual_joint_entropy_after_update; actual_joint_entropy_decomposition`

7 A stationary perturbation of the continuous-boundary reference

We now use one genuine continuous boundary coordinate, so I has one element and $\mathsf{P} = G$. Every finite word collection and finite real coupling in (9) is still permitted. Define the open event

$$H = \{U \in G : \operatorname{Re} U_{11} > 0\}.$$

Its complement has a nonempty open subset $H_- = \{U : \operatorname{Re} U_{11} < 0\}$.

Lemma 7.1 (A nontrivial event for the actual boundary)**.** *For every finite word interaction and $\beta \in \mathbb{R}$,*

$$0 < a_\beta := \tau_\beta(H) < 1.$$

Proof. The matrix entry is continuous. The identity belongs to H, and -1 belongs to H_-, so both are nonempty open sets. Every nonempty open set in a compact group has positive Haar measure: if O is nonempty, its left translates cover the group, and compactness extracts a finite subcover. If $m(O) = 0$, translation invariance would make the whole group have measure zero, contradicting $m(G) = 1$. Thus $m(H) > 0$ and $m(H^c) \geq m(H_-) > 0$. Proposition 5.1 says that τ_β has exactly the Haar null sets, so the same strict inequalities hold for τ_β. Since it is a probability and H is measurable, they imply $0 < \tau_\beta(H) < 1$. □

Source correspondence: `SU2BoundaryConditionedEntropy`
`haar_hemisphere_positive; haar_complement_positive`

Source correspondence: `SU2BoundaryConditionedEntropy`
`boundary_null_iff; boundary_event_nontrivial`

No symmetry of the interaction has been assumed, and $a_\beta = 1/2$ is not claimed. The argument proves the strict range actually required for finite positive entropy.

Theorem 7.2 (Continuous-boundary entropy obstruction)**.** *Define*

$$\widehat{\tau}_\beta = \tau_\beta(\,\cdot \mid H), \qquad \widehat{Q}_\beta = \widehat{\tau}_\beta \otimes K_\beta.$$

These are probability measures, $\widehat{Q}_\beta$ is stationary for T_β, and

$$0 < \mathrm{D}(\widehat{Q}_\beta \| Q_\beta) = \log(1/a_\beta) < \infty. \tag{11}$$

For every $q < 1$ the strict global contraction inequality

$$\mathrm{D}(JT_\beta \| Q_\beta) \leq q\, \mathrm{D}(J \| Q_\beta)$$

fails for the actual finite-entropy input $J = \widehat{Q}_\beta$.

Proof. Lemma 7.1 makes conditioning well defined. Lemma 4.1, applied to τ_β, K_β, H, gives stationarity and (11). Substitution into the proposed inequality gives $\log(1/a_\beta) \leq q \log(1/a_\beta)$, impossible for $q < 1$. □

Source correspondence: `SU2BoundaryConditionedEntropy`
`perturbed_stationary; entropy_finite; entropy_exact`

Source correspondence: `SU2BoundaryConditionedEntropy`
`entropy_positive; no_strict_contraction`

This test changes the boundary distribution absolutely continuously and retains the correct link conditional at every retained boundary. It is therefore stronger for this particular reference-law question than changing the reference to a two-atom boundary law. The two examples share the same classical mechanism, but they address different reference measures.

8 A replacement kernel with genuine entropy decay

For comparison we now change the state space and the update. The state is one loop matrix $C \in G$. Put $h(C) = \frac{1}{2} \operatorname{Re} \operatorname{tr} C$, so $|h(C)| \leq 1$. For $r \in \mathbb{R}$, define

$$z_r(C) = \int_G e^{rh(C)h(D)} m(dD), \qquad Z_r = \int_G z_r(C) m(dC),$$

$$k_r(C,D) = \frac{e^{rh(C)h(D)}}{z_r(C)}, \quad P_r(C,dD) = k_r(C,D)m(dD),$$

$$\pi_r(dC) = p_r(C)m(dC), \quad p_r(C) = \frac{z_r(C)}{Z_r}.$$

This transition outputs D as the next state; it does not output the pair (C,D) while retaining C. It is therefore different from the update T on a boundary–interior pair. All section and total partitions are between $e^{-|r|}$ and $e^{|r|}$, so the formulas give actual probabilities and a Markov kernel by the same measurable-density argument as before.

Proposition 8.1 (Stationary comparison bounds)**.** *The law π_r is stationary for P_r, and*

$$e^{-2|r|} \le k_r(C,D), p_r(D) \le e^{2|r|}.$$

In particular, with $\varepsilon_r = \frac{1}{2}e^{-4|r|} \in (0,1)$,

$$\varepsilon_r \pi_r \le P_r(C,\cdot) \le e^{4|r|}\pi_r \qquad \text{for every } C. \tag{12}$$

Proof. The joint density $p_r(C)k_r(C,D)$ equals $Z_r^{-1}e^{rh(C)h(D)}$. It is symmetric in C, D. Its first marginal is π_r, so symmetry makes its second marginal π_r as well. This is stationarity. The two density bounds follow by dividing numerator and partition bounds. Consequently $k_r(C,D) \ge e^{-2|r|} \ge e^{-4|r|}p_r(D)$ and $k_r(C,D) \le e^{2|r|} \le e^{4|r|}p_r(D)$. Integrate these inequalities against m and weaken the first coefficient by a factor two to obtain (12). The factor two ensures $\varepsilon_r < 1$ even at $r = 0$ and matches the source's residual kernel. □

Source correspondence: `SU2LoopEquilibrium`
`detailed_balance_density; equilibriumLaw_stationary`

Source correspondence: `SU2StationaryMinorization`
`stationary_density_minorization; loopKernel_minorizes_equilibrium`

Source correspondence: `SU2PointEntropy`
`reverse_density_bound; loopKernel_dominated_by_equilibrium`

We give the entropy implication with enough detail to specify the actual residual kernel and the role of finite entropy. The following argument is classical. Suppose P is a Markov kernel on a standard Borel space, π is stationary and $P(x,\cdot) \ge \varepsilon\pi$ for every x, with $0 < \varepsilon < 1$. Define

$$R(x,\cdot) = \frac{P(x,\cdot) - \varepsilon\pi}{1-\varepsilon}.$$

This is a Markov kernel: subtraction leaves a nonnegative measure by minorization, its mass is $1-\varepsilon$, and measurable evaluations follow from those of P. Integrating the identity $P = \varepsilon\pi + (1-\varepsilon)R$ against π and using stationarity gives $\pi R = \pi$.

Lemma 8.2 (Residual entropy contraction)**.** *For every probability μ with finite* $\mathrm{D}(\mu\|\pi)$*,*

$$\mathrm{D}(\mu P\|\pi) \le (1-\varepsilon)\,\mathrm{D}(\mu\|\pi).$$

Proof. We first verify data processing for R in this finite-entropy setting. Write $\mu = f\pi$. The stationary joint law $\pi(dx)R(x,dy)$ has second marginal $\pi(dy)$. Its disintegration given y supplies

a reverse kernel $\widetilde{R}(y, dx)$. The density of μR relative to π is $g(y) = \int f(x)\widetilde{R}(y, dx)$, as checked by integrating against bounded measurable functions of y. Jensen for $u \log u$ gives

$$\int g \log g \, d\pi \le \int\int f(x) \log f(x) \widetilde{R}(y, dx)\pi(dy) = \int f \log f \, d\pi.$$

Integrable negative parts justify these extended integrals; the upper bound is finite, so $\mathrm{D}(\mu R\|\pi)$ is finite too.

The density of μP is $\varepsilon + (1-\varepsilon)g$. Convexity of $u \log u$ at the mixture of 1 and g gives pointwise

$$(\varepsilon + (1-\varepsilon)g) \log(\varepsilon + (1-\varepsilon)g) \le (1-\varepsilon)g \log g.$$

Integration and the data-processing bound prove the claim. □

Source correspondence: `SU2ResidualEntropy`
`residualKernel_markov; residualKernel_split; residualKernel_invariant`

Source correspondence: `SU2ResidualEntropy`
`entropy_contraction; entropy_geometric`

The mapped formal theorem also allows infinite input entropy using extended KL inequalities. The finite-input proof above suffices after the following actual smoothing step and does not rely on multiplying an infinite entropy by a geometric factor and then calling it a convergence estimate.

Theorem 8.3 (Finite entropy after one step and subsequent decay)**.** *For every $r \in \mathbb{R}$ and* every *probability input μ on G, including a point mass,*

$$\mathrm{D}(\mu P_r\|\pi_r) \le 4|r| < \infty,$$

and for all integers $n \ge 0$,

$$\mathrm{D}(\mu P_r^{n+1}\|\pi_r) \le (1-\varepsilon_r)^n \, \mathrm{D}(\mu P_r\|\pi_r) \le 4|r|(1-\varepsilon_r)^n \longrightarrow 0. \tag{13}$$

Proof. Integrating the upper bound in (12) against μ gives $\mu P_r \le M\pi_r$, $M = e^{4|r|} \ge 1$. Its density f is at most M. Since $f \log f \le f \log M$, its entropy is at most $\log M = 4|r|$; the negative part is bounded, so this also proves finiteness. Now apply Lemma 8.2 successively to the finite-entropy input μP_r. Every iterate has finite entropy by induction, giving the first inequality in (13). The geometric factor tends to zero because $0 < \varepsilon_r < 1$. This proves the last assertion, including $r = 0$, where the first-step entropy is zero. □

Source correspondence: `SU2PointEntropy`
`forward_entropy_finite; entropy_geometric_after_first`

Source correspondence: `SU2PointEntropy`
`point_start_entropy_finite; entropy_tendsto_zero_after_first`

The explicit upper bound $4|r|$ is a printed human consequence of the mapped domination estimate; the source states finiteness and the exact geometric estimate with the actual first-step entropy. No sharpness of the numerical constant is claimed. The rate here is fixed-coupling: the displayed ε_r tends to zero as $|r| \to \infty$, and this proof supplies no coupling-uniform entropy rate. Neither it nor the boundary obstruction addresses volume or continuum uniformity.

9 A two-step Haar reset through a zero-trace band

The replacement kernel has a more informative positive-coupling reset than its global density bound alone suggests. Keep the *same* P_r from the preceding section, and assume $r \geq 0$. For $\delta > 0$ put

$$B_\delta = \{Z \in G : |h(Z)| < \delta\}, \qquad a_\delta = m(B_\delta), \qquad b_{r,\delta} = a_\delta e^{-(1+3\delta)r}.$$

The set is open since h is continuous. It is nonempty, since $\mathrm{diag}(i, -i) \in \mathrm{SU}(2)$ has zero trace. Thus $a_\delta > 0$. For completeness, Haar probability on a compact group assigns positive mass to every nonempty open set: its left translates cover the group, a finite subcover exists by compactness, and zero mass of the original set would force total mass zero. Hence no unproved quantitative band mass estimate is needed here.

Theorem 9.1 (Actual two-step minorization)**.** *For every $C \in G$,*

$$P_r^2(C, \cdot) \geq b_{r,\delta} m, \qquad 0 < b_{r,\delta} \leq 1. \tag{14}$$

The composition is of the actual probability kernel, twice; it is not a proposed transition density or a stationary-reference reset.

Proof. For $Z \in B_\delta$, the first density obeys

$$k_r(C, Z) \geq e^{-(1+\delta)r},$$

because $rh(C)h(Z) \geq -r\delta$ and $z_r(C) \leq e^r$. At the second step, $z_r(Z) \leq e^{r\delta}$ and $rh(Z)h(D) \geq -r\delta$ for every D. Thus

$$k_r(Z, D) \geq e^{-2r\delta} \qquad (Z \in B_\delta).$$

For any Borel A, positivity and Tonelli give

$$\begin{aligned} P_r^2(C, A) &= \int_G k_r(C, Z) \int_A k_r(Z, D) m(dD) m(dZ) \\ &\geq \int_{B_\delta} e^{-(1+\delta)r} e^{-2r\delta} m(A) m(dZ) = b_{r,\delta} m(A). \end{aligned}$$

The lower coefficient is positive. Evaluating the inequality at $A = G$ gives $b_{r,\delta} \leq 1$, using the Markov property of the composition. □

Source correspondence: `SU2ZeroTraceBridge`
`band_mass_pos; density_into_band; density_from_band`

Source correspondence: `SU2ZeroTraceBridge`
`two_step_reset; bridge_pos; bridge_le_one`

Iteration, stationary convergence and the pair count

Put $\alpha = b_{r,\delta}/2 \in (0, 1)$ and define the genuine residual

$$L(C, \cdot) = \frac{P_r^2(C, \cdot) - \alpha m}{1 - \alpha}.$$

Nonnegativity, measurable evaluations and mass one follow just as in the previous residual construction. Here m is a common reset law; it is not asserted stationary. In fact $\pi_r = \alpha m + (1 - \alpha)\pi_r L$

follows from stationarity of π_r for P_r^2, and does not give $\pi_r L = \pi_r$. For a bounded real measurable f, let $\operatorname{osc}(f) = \sup_C f(C) - \inf_C f(C)$. Since a Markov average stays between the infimum and supremum of its input,

$$\operatorname{osc}(P_r^2 f) = (1-\alpha)\operatorname{osc}(Lf) \le (1-\alpha)\operatorname{osc}(f).$$

Induction therefore proves, for $0 \le f \le 1$,

$$|P_r^{2n} f(C) - P_r^{2n} f(D)| \le (1-\alpha)^n \le e^{-n\alpha}. \tag{15}$$

All powers are actual kernel powers. Measurability and boundedness are preserved by kernel integration, so every induction step is defined. The same estimate holds for $2n+1$ steps by applying the even-step bound to $P_r f$, which still takes values in $[0,1]$.

Source correspondence: `SU2ZeroTraceBridge twoAverage_eq_transfer_twice; iterate_range`

Source correspondence: `SU2ZeroTraceBridge iterate_oscillation; iterate_exponential_bound`

A useful human consequence is convergence from every initial probability μ to the already constructed stationary π_r. Define $d_{\mathrm{TV}}(\mu,\nu) = \sup_A |\mu(A) - \nu(A)|$. Integrating the two-state bound for $f = 1_A$ against $\mu(dC)\pi_r(dD)$ and using stationarity gives

$$d_{\mathrm{TV}}(\mu P_r^{2n}, \pi_r), \quad d_{\mathrm{TV}}(\mu P_r^{2n+1}, \pi_r) \le (1-\alpha)^n.$$

This also proves uniqueness: if ν is stationary, substitute ν and let $n \to \infty$. This total-variation conclusion is a printed human deduction, not a new mapped formal theorem.

For a prescribed $t \ge 0$, a sufficient integer pair count is

$$n \ge \frac{2t}{a_\delta} e^{(1+3\delta)r}. \tag{16}$$

Multiplication by $\alpha = (a_\delta/2)e^{-(1+3\delta)r}$ gives $n\alpha \ge t$, and (15) then gives e^{-t}. These are n pairs, hence $2n$ original transitions. For each $\eta > 0$, choose $\delta = \eta/3$; the positive two-step Haar coefficient is then $a_{\eta/3}e^{-(1+\eta)r}$. Its prefactor depends on η. No exponent-one endpoint, optimality or coupling-uniform rate is claimed.

Source correspondence: `SU2ZeroTraceBridge sufficient_pair_count; reset_rate_near_one`

The oscillation/total-variation mechanism is classical Doeblin contraction; Gaubert–Qu [3], introductory equations (1)–(4) and Theorem 1.1, give its coefficient and oscillation context. No new general ergodicity principle is asserted. This Haar minorization is also not a license to reuse Lemma 8.2 with reference π_r: that entropy proof required a *stationary-reference* minorization and invariant residual. The earlier one-step entropy conclusion retains its separate proof.

The zero-trace row distinguishes Wilson transport

There is a direct obstruction to identifying this reset mechanism with one-word Wilson transport. Define the latter on the same group by

$$W_r(C, dD) = \frac{e^{rh(C^{-1}D)}}{\zeta_r} m(dD), \qquad \zeta_r = \int_G e^{rh(V)} m(dV).$$

Left invariance makes the denominator independent of C. Equivalently, $D = CV$, where V has the Haar tilt $e^{rh(V)}/\zeta_r$. This is the source's one-word transition evaluated at the inverse boundary argument. Its normalized relative-holonomy mean is

$$\int_G h(C^{-1}D)W_r(C, dD) = c_r, \qquad c_r = \frac{\int_G h(V)e^{rh(V)}m(dV)}{\zeta_r}.$$

In contrast, $h(C) = 0$ makes $P_r(C, \cdot) = m$ exactly, since its interaction vanishes and its partition is one.

Source correspondence: `SU2ZeroTraceBridge zero_trace_row; wilson_relative_mean`

Here is a direct human strictness argument, valid at every $r > 0$. The central transformation $V \mapsto -V$ preserves Haar probability and sends h to $-h$. Averaging the original numerator with its transformed expression gives

$$\int h(V)e^{rh(V)}m(dV) = \int h(V)\sinh(rh(V))m(dV) > 0.$$

The integrand is nonnegative and continuous, and strictly positive on a nonempty open neighborhood of the identity. The same compact Haar support argument gives the strict integral inequality. All integrands are bounded at each fixed r. Thus $c_r > 0$, whereas $\int h(C^{-1}D)m(dD) = \int h\,dm = 0$ by the same sign symmetry. At any zero-trace C the two rows consequently differ for every $r > 0$. The mapped formal mismatch is stated eventually at large coupling, using $c_r \to 1$; the all-positive-coupling strengthening here is a labelled human symmetry proof, not that literal formal quantifier.

Source correspondence: `SU2ZeroTraceBridge zero_trace_wilson_mismatch`

This distinction already occurs for one group variable. It does not settle a Wilson lattice gap or prohibit another valid minorization for W_r. In particular the algebraic rescaling $2ne^{-(1+3\delta)r}$ in the loop oscillation exponent is only a proposed distance scale until related independently to an actual lattice spacing. The source labels that scale as proposed; no physical mass gap follows by renaming a transition count.

10 Interpretation and comparison with heat-bath mixing

Conditional expectation and the entropy chain rule are classical. This paper makes no novelty claim for the obstruction that an exactly retained variable can preserve information. Raginsky's primary treatment of strong data processing [1], Definition 1, formulates strict entropy contraction as a ratio bounded below one. The stationary tests above exhibit ratio one in the specified channel. His finite-alphabet results are background for the terminology; we do not substitute them for the continuous-boundary proof.

Caputo–Menz–Tetali [2], Section 1.1 and Proposition 1.1, treat heat-bath dynamics that update all coordinate positions, with approximate entropy tensorization supplying global control. That is a different update from repeating one boundary-preserving conditional step. Their weak dependence hypotheses are not established here, and no such mixing estimate is imported. Our stationary family does not contradict entropy decay for a dynamics whose allowed updates also change the boundary.

There is also no conflict with strict mixing of a finite transfer kernel on a different state space. The finite seven-link transfer construction in the companion paper updates its own chain state; it is

not identified here with T_β. A kernel preserving U exactly is structurally incapable of forgetting an arbitrary initial distribution of U, regardless of how positive its conditional link density is. Boundary reweighting in (10) specifies which global reference is at issue.

For the restricted class $J_{\mathsf{P}} = \tau_\beta$, formula (2) sends J to Q_β in one step. For arbitrary boundary inputs the global fraction fails at every fixed finite coupling, rather than only in a large-coupling asymptotic. These two exact statements delimit the scope without claiming an entropy inequality that the update cannot satisfy.

Finally the hemisphere event uses a particular matrix coordinate and is not asserted gauge invariant. The theorem concerns the stated actual probability space and conditional update; it does not classify a gauge quotient, all lattice block dynamics, infinite-volume measures or continuum Schwinger functions. No lattice-spacing or volume limit is constructed, and no physical mass-gap implication, positive or negative, follows.

References

[1] M. Raginsky, *Logarithmic Sobolev inequalities and strong data processing theorems for discrete channels*, IEEE International Symposium on Information Theory (2013). Primary author PDF.

[2] P. Caputo, G. Menz and P. Tetali, *Approximate tensorization of entropy at high temperature*, Ann. Fac. Sci. Toulouse Math. (6) **24** (2015), 691–716. Primary journal PDF.

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